Alexandria West Wastewater Treatment Plant

# Planning package

Underground pipe works sequence, methodology and programme · Whole-project estimate BOQ

Issue date 22 October 2026 · Prepared by Claude for Mohamed (Projects Controls Director)

Status: issued for information and planning. Not a design, temporary-works design or construction approval.

**Contents**


1. Executive summary
2. Document register
3. Basis and decisions
4. Groundwater and temporary works
5. Construction methodology
6. Programme
7. Aeration piling status
8. Estimate BOQ
9. Open items, exclusions and how to use the files


Inputs: pipe execution packages v3 (21 Oct 2026), drawing ALEXW-MWT-MEC-00-DLU-002 Rev 03 (Issued for Approval), combined coordination model (19 Oct 2026), decisions of 20-22 Oct 2026.

## 1. Executive summary

| 9.6 months | 80% | 145,800 m³ | 16,000 |
|---|---|---|---|
| Pipe works, 1 Dec 2026 to 18 Sep 2027 (best case, gates not yet entered) | Pipe length in sheet piles with groundwater at -1.50 (was 24%) | Reinforced concrete, whole project (gross) | Piles 600 mm (model, allowance and pipe placeholder) |

- **Groundwater at -1.50 changes the pipe works.** 199 of 201 packages excavate below groundwater, by up to 6.1 m. With the assumed limit of 2.5 m below GWL for open cut, 6,562 m (80%) of the 8,151 m of pipe needs sheet piles, about 116,300 m² of sheets, against 1,972 m before groundwater was considered.
- **Every pipe sits on piles and a piled slab.** That is about 5,800 pipe-support piles on a placeholder layout. Piling is the longest activity on most fronts and peaks at about 13 rigs in December 2026.
- **Aeration rigs can hand over.** 8 rigs are completing aeration piling at a measured 5.4 piles per rig per day; the forecast is end November 2026, just as pipe piling starts.
- **Flotation.** 14 packages on UG-08, UG-09 and UG-10 (DN2600, shallow cover) can float even at final level and need permanent straps to the piled slab. A further 72 need dewatering kept on until the fill reaches finished level.
- **Programme.** Eight fronts and 2,211 logic-linked activities give 1 December 2026 to 18 September 2027. This is a best case: approval gates and structure readiness dates are not yet entered.
- **Whole-project quantities.** About 145,800 m³ reinforced concrete, 17,300 t reinforcement, 221,000 m² formwork, 16,000 piles and 8,294 m of pipe, all from the 3D model and the pipe workbooks. The aeration raft reconciles to the aeration quantity file within 0.4%. The BOQ is unpriced, and structure excavation is still overstated.

> **Decisions needed to firm this up:** confirm the groundwater design value and the open-cut rule (geotechnical); pipe and structure pile designs; structure readiness dates for tie-ins; DLU-002 approval forecast; unit rates.

## 2. Document register

All files are in the package folder and on branch `claude/new-session-e5ojdh` of repository engmohamedshipl-arch/claude.

| No. | Title | File | Format | Status |
|---|---|---|---|---|
| PKG-00 | Planning package report (this document) | `Alex-West-Planning-Package-Report.pdf` | PDF | Issued for information |
| PKG-01 | Pipe sequence and methodology workbook | `01-Sequence/Alex-West-Pipe-Sequence-Methodology.xlsx` | Excel, live formulas | Planning draft |
| PKG-02 | Pipe works programme for Primavera P6 | `02-Programme/Alex-West-Pipe-Works.xer` | P6 XER | Planning draft |
| PKG-03 | Pipe works programme for MS Project | `02-Programme/Alex-West-Pipe-Works.xml` | MSPDI XML | Planning draft |
| PKG-04 | Estimate bill of quantities, whole project | `03-BOQ/Alex-West-Estimate-BOQ.xlsx` | Excel, live formulas | Quantities only, unpriced |
| PKG-05 | Project wiki (decisions, open items, method notes) | `04-Wiki/*.md` | Markdown | Living record |
| PKG-06 | Build scripts and extracted model data | `05-Scripts-and-Data/` | Python, JSON | For rebuild |
| REF-01 | Pipe execution packages v3 (input, 21 Oct 2026) | `not included - project file` | Excel | Input |
| REF-02 | Combined coordination model (input, 19 Oct 2026) | `not included - project file (164 MB)` | Blender | Input |

### 3. Basis and decisions

| Ref | Date | Decision | Owner |
|---|---|---|---|
| D-001 | before 20 Oct | Excavation bottom = external pipe bottom - 0.50 m (piled slab) | Mohamed |
| D-004 | 20 Oct | NG -0.70, FL +1.00, minimum cover 1.0 m | Mohamed |
| - | 21 Oct | Working space 1.0 m; no maximum package length; shoring where open cut does not fit | Mohamed |
| - | 22 Oct | Groundwater -1.50; all pipes on piles and piled slab; shoring as needed | Mohamed |
| - | 22 Oct | Thickener bridge carrying the mechanism is reinforced concrete | Mohamed |
| - | 18 Oct | Aeration piling order: zones 1, 2, 3, then 5+4, then 6, 7, 8 | Agreed timeline |

Assumptions by Claude (slopes 1.5 and 2.0 H:V, open-cut limit 2.5 m below GWL, pile layout, rebar ratios, productivity) are listed on each workbook's Inputs sheet and in the wiki decisions log.

## 4. Groundwater and temporary works

Sheet-piled length of the 8,151 m pipe network under alternative rules. The highlighted bar is the default in the workbooks. The rule is the single largest driver of temporary-works cost.

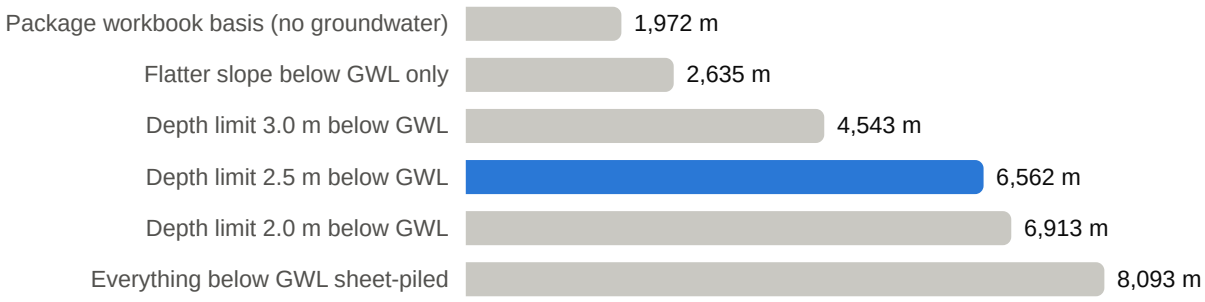



Figure 1 - Sheet-piled pipe length by rule (m). Source: sequence workbook, Sensitivity sheet.

| Topic | Result |
|---|---|
| Dewatering | 160 packages single-stage wellpoints; 39 deep wells or two-stage wellpoints (drawdown beyond 4.5 m); 2 not required |
| Just-in-time pumping | Drawdown timed to finish when the excavation crew is free; median 74 pumping days per package |
| Flotation, final | 14 packages (UG-08, UG-09, UG-10): permanent anchor straps to the piled slab |
| Flotation, construction | 72 packages: keep dewatering until fill reaches FL, or strap to slab |
| Sheet piles | About 116,300 m² (wall length x 2 x excavation depth); temporary, extracted after backfill |

## 5. Construction methodology (per package)

| Step | Activity | Method and hold points |
|---|---|---|
| 1 | **Release** | DLU-002 approved, pile design approved, structure ready (tie-ins), aeration zone piling complete (within 15 m), crossings identified |
| 2 | **Set-out and platform** | Survey axis, piles, sheet lines; trial pits at crossings; granular platform at NG for rigs |
| 3 | **Pipe-support piling** | From the platform before excavation; 600 mm assumed; integrity tests; preliminary load test |
| 4 | **Sheet piles** | Where open cut does not fit or excavation is more than 2.5 m below GWL; press-in near structures |
| 5 | **Dewatering** | Wellpoints, or deep wells beyond 4.5 m drawdown; at least 7 days drawdown; piezometers and settlement markers |
| 6 | **Excavate and strut** | In layers, evenly around installed piles; deeper lines before shallower; shared trenches opened together |
| 7 | **Piled slab** | Trim piles, blinding, 0.50 m slab with starter bars; straps cast in where flotation requires |
| 8 | **Lay and joint** | Downstream first; GRP bell and spigot, HDPE butt fusion; tie-ins last as closure |
| 9 | **Surround and test** | Granular haunch and surround; joint and line tests |
| 10 | **Backfill** | Layers to NG, struts out progressively; dewatering kept on until the flotation check is met |
| 11 | **Off and extract** | Dewatering off, sheets extracted slowly, voids grouted, surface reinstated |

### Sequencing rules

- Eight fronts by system. A shared-trench corridor that crosses systems is built by one front.
- Within a front: deeper lines first, downstream first, shared-trench packages back to back.
- Deeper-first links where packages are within 8 m and bottoms differ by more than 1 m. Shared trenches (bottoms within 1 m) are opened together and backfilled after all pipes are laid (6 holds are site coordination items).
- Tie-ins follow the adjacent run, wait for structure readiness and close the line.
- Packages within 15 m of open aeration piles wait for that zone. Pipe piling waits for pile design approval; nothing starts before DLU-002 approval.

## 6. Programme

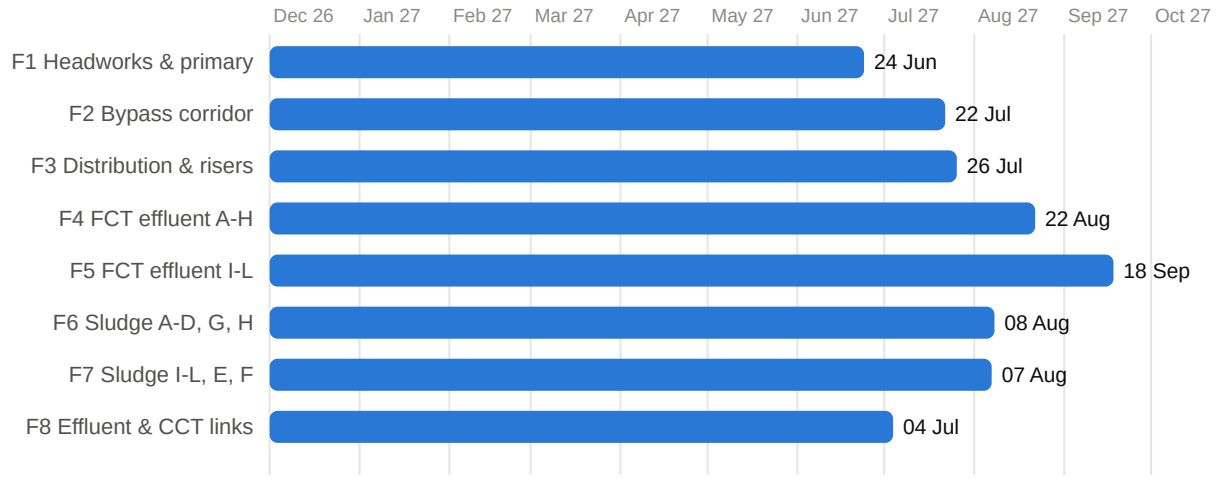



Figure 2 - Pipe works by front (start 1 Dec 2026, all fronts). Source: sequence workbook; same logic in the P6 and MS Project files.

| Front | Scope | Lines | Length (m) | Sheet-piled (m) | Crews | Finish |
|---|---|---|---|---|---|---|
| **F1** | Headworks & primary (GRT-PST-IPS) | UG-01, UG-02, UG-03, UG-04 | 800 | 800 | 1 | 24 Jun 2027 |
| **F2** | Bypass corridor: IPS bypass and outfall twin | UG-05, UG-06, UG-24, UG-25, UG-26 | 1,724 | 1,567 | 2 | 22 Jul 2027 |
| **F3** | Distribution & FCT risers | UG-08, UG-09, UG-10, UG-11B, UG-13A, UG-13B, UG-14E, UG-14F, UG-16B, UG-27J, UG-27K, UG-27L | 1,095 | 649 | 1 | 26 Jul 2027 |
| **F4** | FCT effluent A-H to CCT | UG-07, UG-07A, UG-15, UG-15A | 551 | 308 | 1 | 22 Aug 2027 |
| **F5** | FCT effluent I-L to CCT | UG-07, UG-11, UG-11B, UG-12, UG-12A | 571 | 254 | 1 | 18 Sep 2027 |
| **F6** | Sludge A-D,G,H to RAS/WAS + UG-23 | UG-11, UG-11A, UG-11B, UG-14G, UG-14H, UG-16A, UG-16C, UG-16D, UG-23 | 1,164 | 840 | 1 | 08 Aug 2027 |
| **F7** | Sludge I-L,E,F to RAS/WAS | UG-07A, UG-12, UG-13, UG-13A, UG-13B, UG-23, UG-27I | 868 | 816 | 1 | 07 Aug 2027 |
| **F8** | Effluent, outfall & CCT links | UG-11B, UG-15, UG-15A, UG-17, UG-18, UG-19, UG-20, UG-21, UG-22, UG-28, UG-28A | 1,378 | 1,326 | 2 | 04 Jul 2027 |

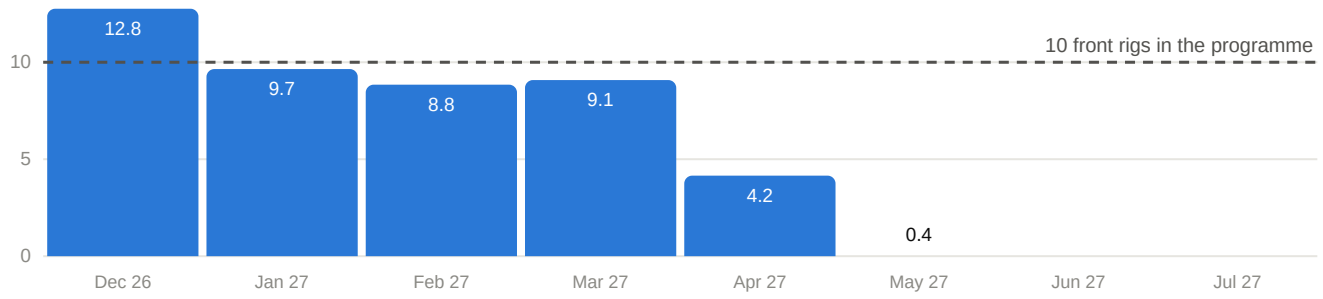



Figure 3 - Equivalent piling rigs needed per month for pipe-support piles (piles per month / 5 per rig per day / 26 days). Demand above the dashed line is tie-in piling, which no front rig covers while structure dates are blank.

## 7. Aeration piling status (model, data date 17 Oct 2026)

| Zone | Done | Pending | Defect | Unknown |
|---|---|---|---|---|
| 1 | 470 | 6 | 2 | 0 |
| 2 | 518 | 41 | 2 | 3 |
| 3 | 248 | 236 | 2 | 0 |
| 4 | 221 | 337 | 3 | 0 |
| 5 | 223 | 137 | 4 | 0 |
| 6 | 134 | 279 | 5 | 0 |
| 7 | 121 | 296 | 1 | 0 |
| 8 | 144 | 236 | 0 | 3 |
| **Total** | **2,079** | **1,568** | **19** | **6** |

"Done" means recorded execution, not acceptance. Over the last 4 full weeks 8-9 rigs averaged 285 piles a week, about 5.4 piles per rig per working day. In the agreed zone order the 1,587 open piles finish around 29 November 2026. That forecast gates the pipe packages within 15 m of open aeration piles and frees the rigs for pipe piling.

## 8. Estimate BOQ (whole project)

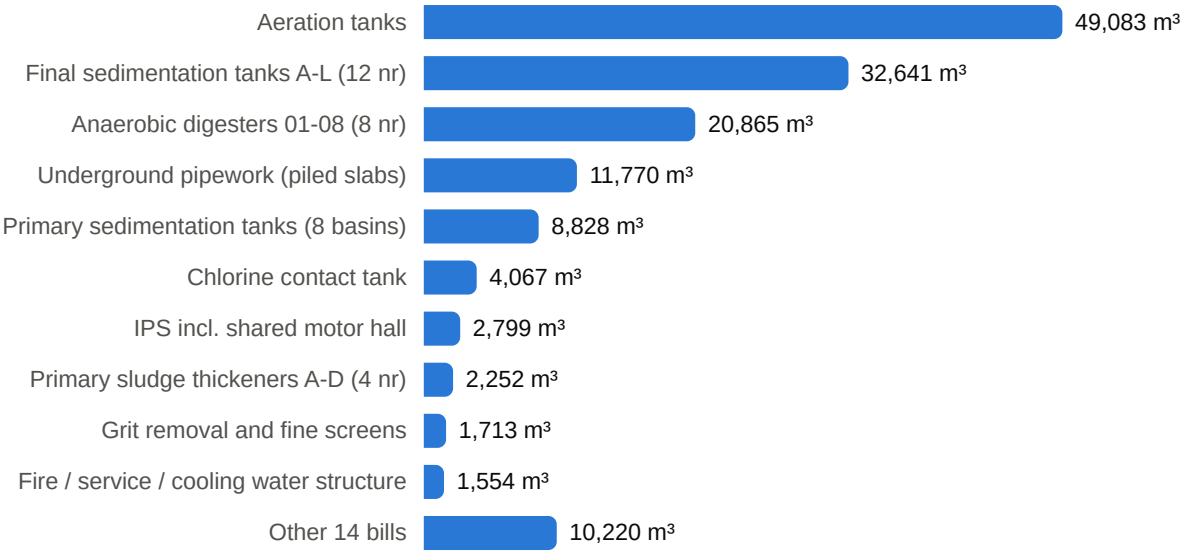



Figure 4 - Reinforced concrete by bill (m³). Source: estimate BOQ, measured on closed 3D model solids.

### Key quantities (gross, no deduction for work done)

| Item | Quantity | Unit |
|---|---|---|
| All reinforced concrete | 145,791 | m³ |
| Reinforcement (ratio basis) | 17,293 | t |
| Formwork | 221,442 | m² |
| Blinding | 6,092 | m³ |
| Piles 600 mm (6,392 model, ~3,800 allowance, 5,788 pipe placeholder) | 15,994 | no. |
| Excavation (structures overstated) | 511,247 | m³ |
| Backfill | 184,618 | m³ |
| Temporary sheet piles (pipes) | 116,309 | m² |
| Pipes | 8,294 | m |
| Pipe fittings | 474 | no. |
| Handrails | 5,412 | m |
| Site raising NG to FL (provisional) | 378,367 | m³ |

- **Cross-checks:** aeration raft 30,120 m³ in the model against 30,013 m³ in the aeration quantity file (0.4%); aeration piles 3,672, the same as the tracking sheet.
- **Confidence** is graded on each BOQ line: B = detailed model; C = civil-guide envelope; D = tender envelope or allowance (digesters, gas holders, sludge buildings, CHP, THP and others).
- **Weak points:** structure excavation is overstated; the thickener bridge is concrete by decision, but the model members are slender (21 m³ in total); only the raft rebar ratio is evidence-based; pile lengths are unknown.
- **Rates** are not included. Enter them in the yellow column and the summary prices itself.

## 9. Open items

| Owner | Item | Affects |
|---|---|---|
| Geotechnical / temporary works | Confirm GWL -1.50, side slopes and the 2.5 m open-cut depth limit below GWL | Sheet-pile quantity swings from 24% to 99% of length |
| Structural design | Pile and slab design for pipe supports; pile layouts for structures without them; thickener bridge sections | 5,788 placeholder and about 3,800 allowance piles; bridge concrete |
| Master programme | Date each structure is ready for pipe tie-in | 71 tie-in packages |
| Design management | Forecast approval of DLU-002 and of the pipe pile design | Gates on every package and all piling |
| Survey / design | Identify crossing pipes P0114, P0117, P0279; surveyed site boundary and levels | 3 packages; provisional site raising |
| Construction / QS | Productivity rates (all assumed except piling); unit rates for the BOQ | Durations; BOQ amounts |
| Planning (Claude) | Refine structure excavation with a grid of each underside level (current method overstates PST, CCT, RAS/WAS) | BOQ excavation and backfill |

## Exclusions and how to use the files

### Not included

- Preliminaries, contingency, M&E and process equipment, architecture and finishes, roads, drainage, fencing and utilities.
- Temporary works and dewatering for structures (pipe temporary works are included).
- Existing structures and boundaries not in the model: existing PST, bypass and outfall chambers, treated effluent pump station, primary and thickened sludge pump stations, sludge flowmeter chambers.

### Using the files

- **Workbooks:** yellow cells are inputs. Enter approval dates (Inputs), structure readiness (Structures), crews (Fronts) and rates (BOQ); everything recalculates.
- **P6:** File > Import > XER as a new project, then F9. Calendar Sat-Thu 8 h; WBS Project > Front > Line > Package; SS+FF ladders kept.
- **MS Project:** open the XML. One link per task pair (the driving link is kept).
- **Keeping in step:** after changing inputs, re-export the programme files with `export_schedule.py` (see the folder README).
- **Wiki:** the decisions log and open items are the living record. Copy the Markdown pages into the repository's GitHub Wiki tab when it is enabled.

> Common vertical datum not verified (R001). Levels are envelope levels, not inverts (R002), and are interpolated between drawing anchors (R003). DLU-002 Rev 03 is Issued for Approval. These holds apply to every pipe package.